



Cosmology

The study of the Universe

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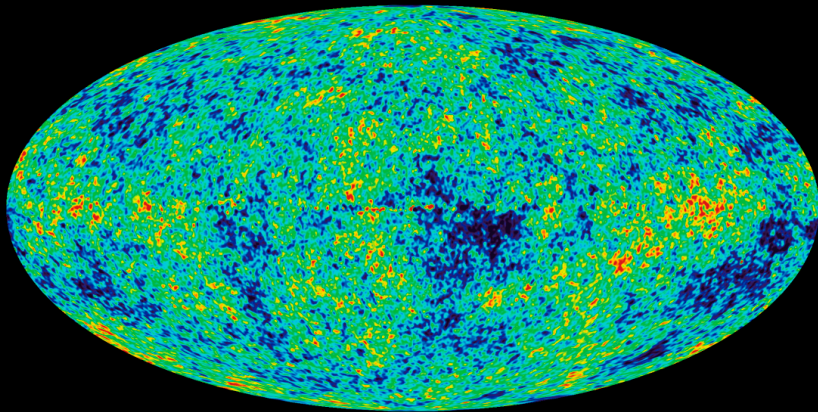
26th September 2012

My life after Alleyn's

- ▶ Went to Gonville & Caius College, Cambridge.
- ▶ Studied Natural Sciences for 4 years.
- ▶ Finished in Experimental & Theoretical Physics.
- ▶ Beginning a PhD in Theoretical Cosmology.



Cosmology



What is Cosmology?

Cosmology is the study of the universe as a whole

- Birth: Is there a beginning of time? What was it like?
- Size: How big is the cosmos? Is there an edge?
- Shape: What happens as you travel across the universe?
- Age: How old is the universe?
- Fate: Will everything come to an end?

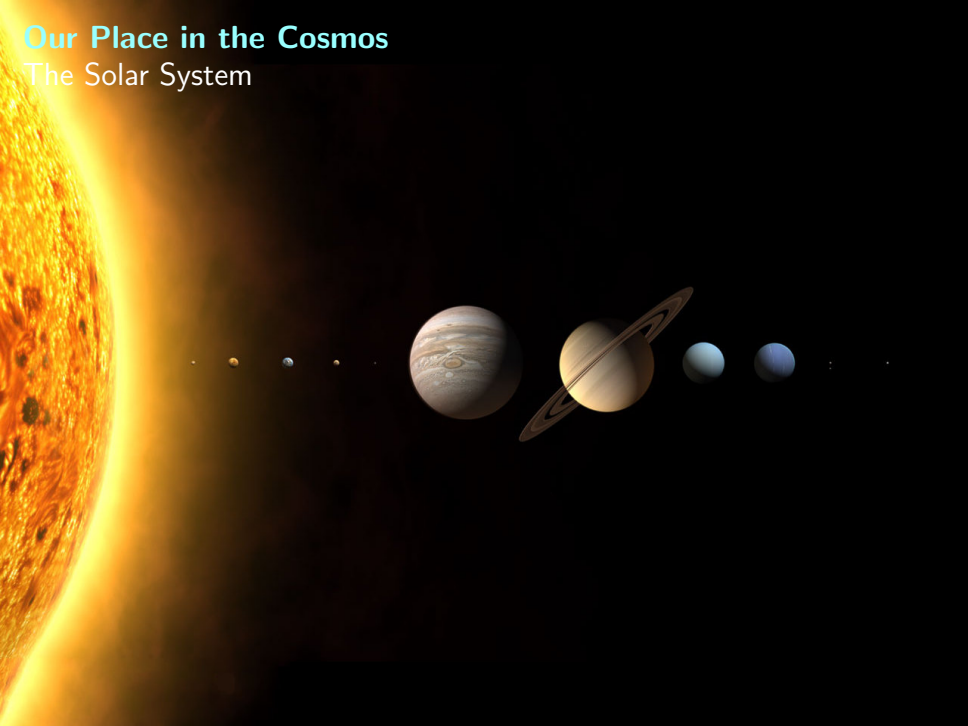
Our Place in the Cosmos

Earth



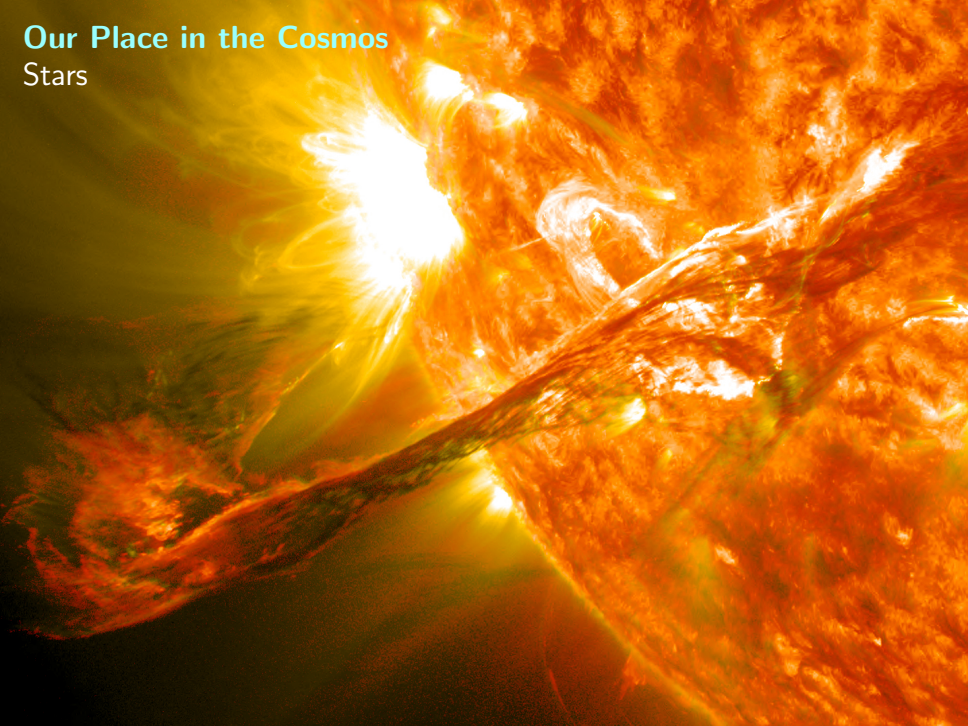
Our Place in the Cosmos

The Solar System



Our Place in the Cosmos

Stars



Our Place in the Cosmos

Star Clusters



Our Place in the Cosmos

Galaxies



Our Place in the Cosmos

Scales in Astronomy

Astronomical Unit (AU):

parsec (pc) / lightyear (ly):

kiloparsec (kpc):

MegaParsec (Mpc):

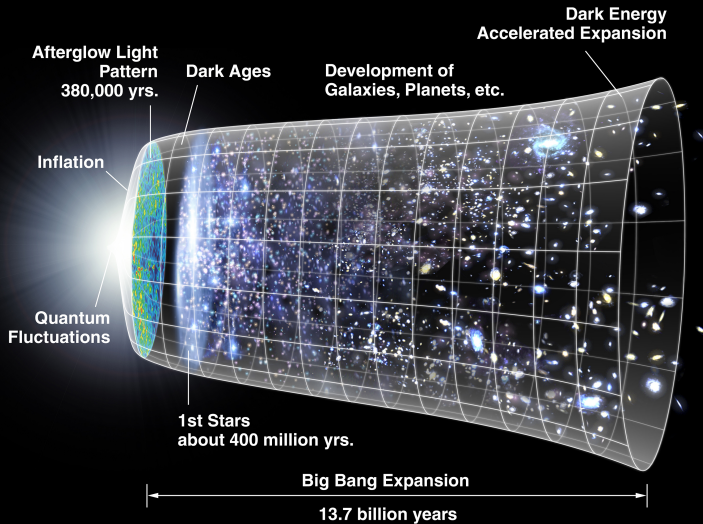
Distance from earth to sun

Distances between stars

Galactic sizes

Galactic separation

The Big Bang



The Big Bang

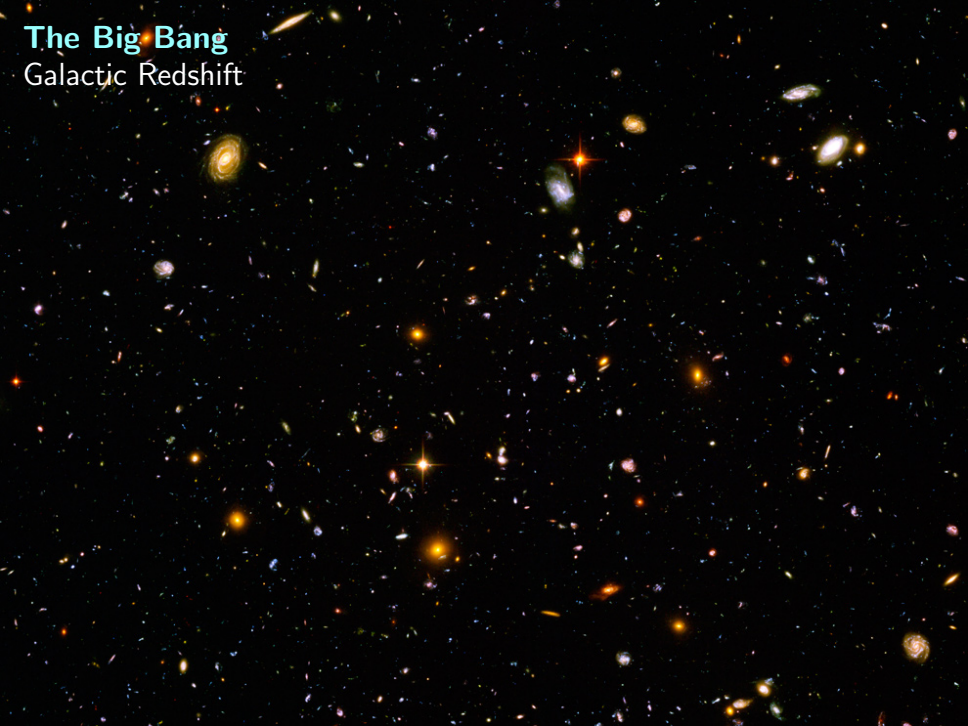
Evidence

There are two fundamental observations which led to the first Big Bang Theory being hypothesised and confirmed:

- ▶ Galactic Redshifts
- ▶ The Cosmic Microwave Background Radiation

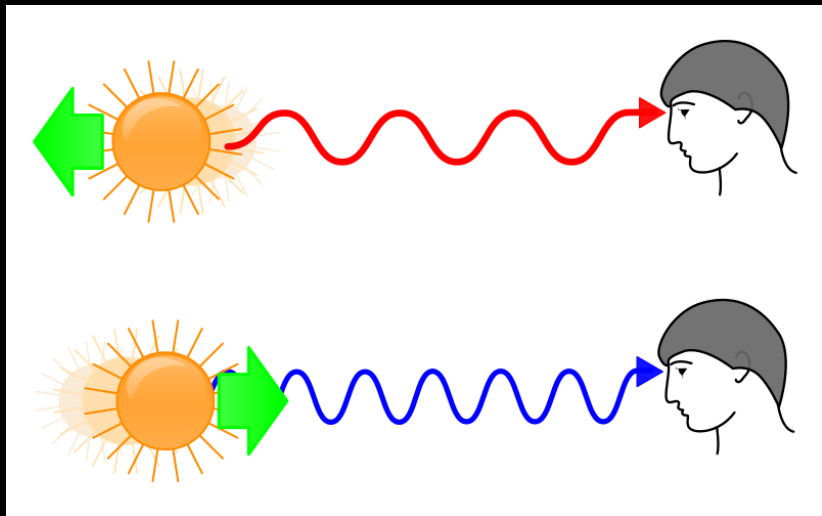
The Big Bang

Galactic Redshift



The Big Bang

Galactic Redshift



The Big Bang

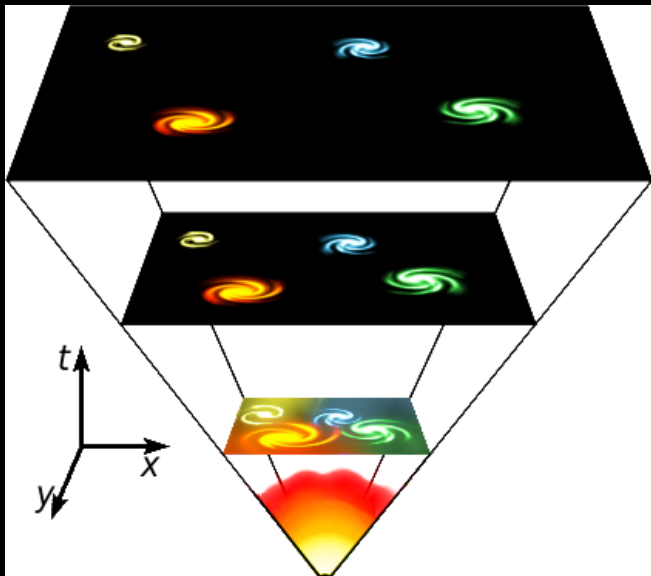
Galactic Redshift

It appears that:

1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away
3. If we were to go to another galaxy, we would see exactly the same thing

The Big Bang

Tracking backward in time...



The Big Bang

The Cosmic Microwave Background Radiation

- ▶ Discovered by accident in 1964 by Penzias & Wilson.
- ▶ Experimenting with a supersensitive detector.



- ▶ Thermal radiation fills the universe at a temperature $T \approx 3K$
- ▶ This is best explained as an 'afterglow' of the Big Bang.

The Big Bang

Common Misconceptions

The Big Bang does *NOT* look like this:



The Big Bang

Common Misconceptions

- ▶ Despite the name, the Big Bang is not an explosion.
- ▶ The Universe is not necessarily expanding into anything.

Constructing a theory of the Universe

Extended Copernican Principle

Philosophically, we expect the universe to be:

Homogeneous: No point in the universe is special.

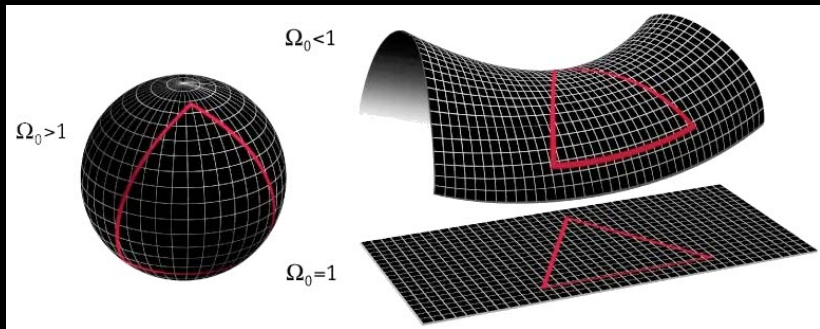
Isotropic: No direction in space is special.

If we demand these conditions we find that there are three types of universe:

- ▶ Open
- ▶ Flat
- ▶ Closed

The Shape, Size & Fate of the Universe

Type	Open	Flat	Closed
Shape	Hyperbolic	Euclidean	Spherical
Size	∞	∞	Finite
Fate	Expanding	Expanding	Eventual Collapse



Current Issues in Cosmology

There are several outstanding problems in Astrophysics which we do not fully understand:

- ▶ Dark Matter
- ▶ Dark Energy
- ▶ $t = 0$

Dark Matter

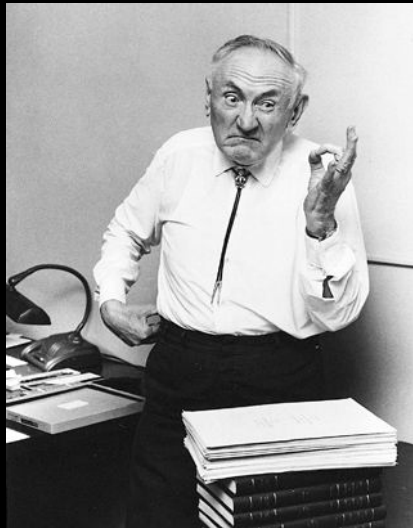
The Problem



Dark Matter

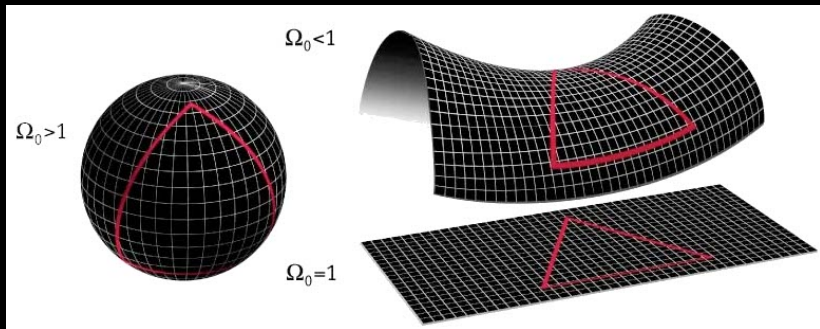
The Problem

- ▶ There appears to be a large amount of matter which we cannot see.
- ▶ There are many other reasons for believing in Dark Matter.
- ▶ Much effort has been put into trying to find the “missing matter” -but so far to no avail.
- ▶ Right now it's really just a name for a phenomenon we don't understand yet.



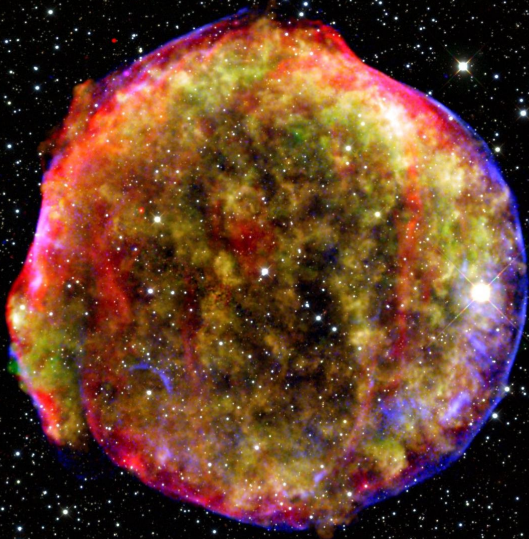
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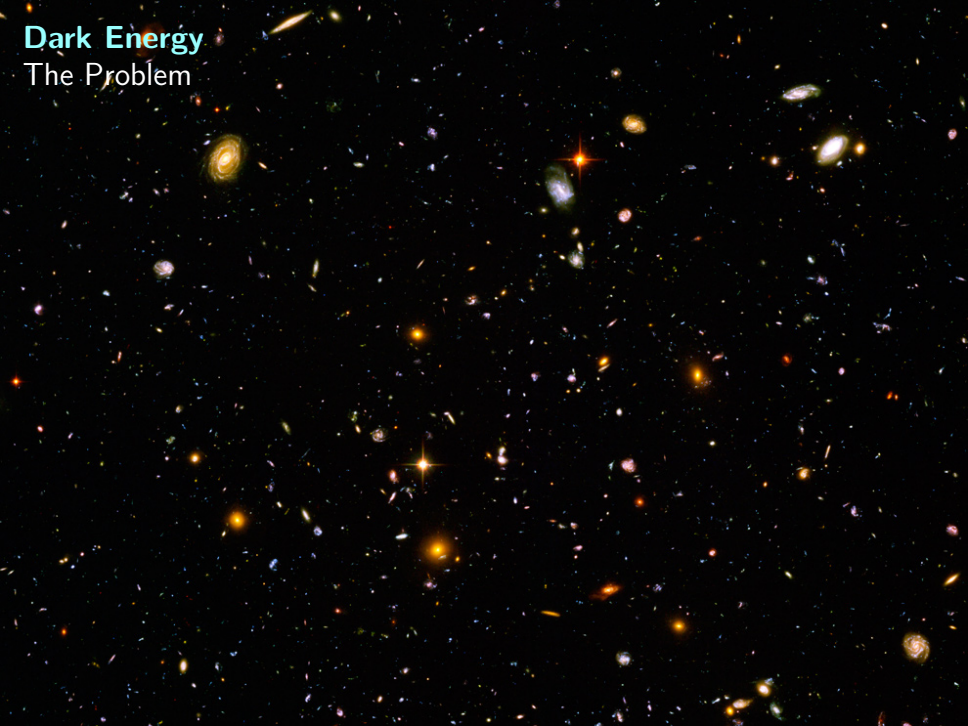
Dark Energy

The Problem



Dark Energy

The Problem

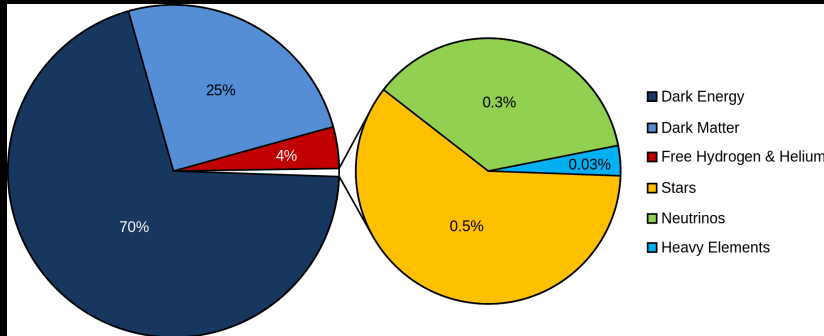


Dark Energy

The Problem

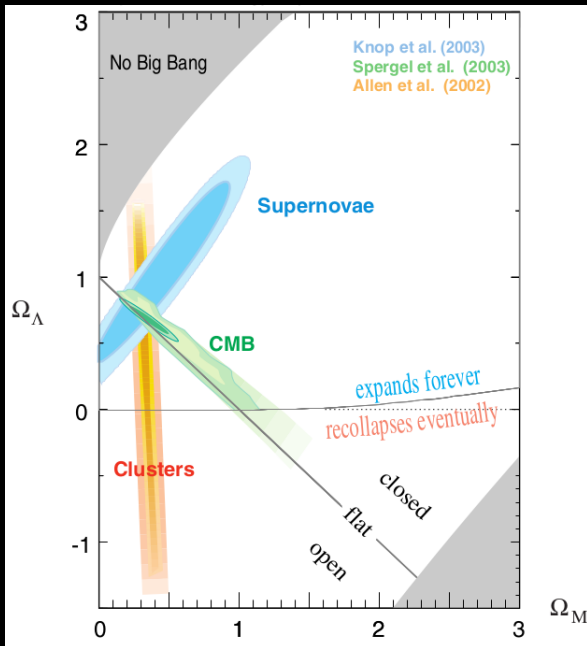
- ▶ The Universe is not slowing down in it's expansion, but *speeding up!*
- ▶ One of the ways in which this could happen would be the presence of an energy permeating the universe, accelerating the expansion.
- ▶ We call this Dark Energy
- ▶ Like Dark Matter, it's really just a name for a phenomenon which we observe, and haven't been able to explain yet

The Composition of the Universe



My Research

- Bayesian Cosmology
- Interface between observation and theory
- Geometric Algebra



Why is Astronomy important?

Why put money into research which is explicitly useless?

Why is Astronomy important?

Why put money into research which is explicitly useless?

1. The techniques which we create in order to study these systems are *very* useful.
2. If there is a theoretical advance in Quantum Gravity, the experimental proof will come from Cosmology.
3. Discovery is important.

Why is Astronomy important?

1. Theoretical & Computational Techniques

- ▶ Massive Projects generate novel ideas in computing, instrumentation and theory.
- ▶ e.g: The Square Kilometer Array
 - ▶ Radio telescope to be built in South Africa and New Zealand.
 - ▶ Will generate PBs of data per second ~ 100 times global internet traffic.



Why is Astronomy important?

1. Theoretical & Computational Techniques

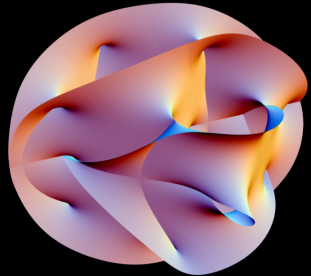
Historically, Astronomy has provided important stepping stones for human progress;

- ▶ Calendars & timekeeping
- ▶ Much of mathematics
 - ▶ Hipparchus → Trigonometry
 - ▶ Newton → Calculus
 - ▶ Logarithms
- ▶ Navigational techniques
- ▶ Driving development of computers
- ▶ Space Age
- ▶ Radio astronomy → communications
- ▶ Optical Astronomy → medical imaging
- ▶ ...

Why is Astronomy important?

2. Quantum Gravity

- ▶ Quantum Mechanics and General Relativity are fundamentally incompatible.
- ▶ To create a unified theory, we should study a system which needs both.
- ▶ The big bang provides an ideal particle accelerator.
- ▶ Cosmology is the perfect playground to test out & discover quantum theories of gravity.



Why is Astronomy important?

3. Discovery

- ▶ The search for new things is part of what makes us fundamentally human.
- ▶ We can't know what's possible until we find it.

